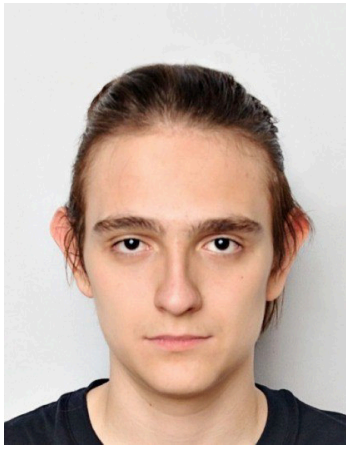




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🌐 d3clane

Education

Moscow State School 57

Physics class, graduated
with gold medal

2019–2023

Moscow Institute of Physics and Technology

Bachelor degree (finished
first year)
Phystech School of Applied
Mathematics and
Informatics

2023–PRESENT

Grades

Overall

Overall GPA: 9.18/10 (5.0/5)

Programming courses
GPA: 9.30/10

Dedinsky' programming courses

First semester: 9/10

Second semester: 10/10

Third semester: 10/10

Ovakimyan Arman

Experience and Projects

AUG 2024

Huawei Open Days

Studied basics of Retrieval-Augmented Generation
(RAG) and its application in AI-based code generation.

DEC 2023 – MAY 2024

Tools: C/C++, ASM X86-64,
Graphviz

X86-64 Compiler for my own programming language

GH link: <https://github.com/d3clane/Compiler>

My own esoteric programming language with compiler
to executable x86-64 elf64 file. Divided into three
programs:

1. Frontend
2. Middle-end
3. Backend

APR 2025

Tools: C/C++, LLVM, Graphviz

LLVM Pass

GH link: https://github.com/d3clane/llvm_pass

Developed an LLVM Pass to gather debug information,
built control-flow and def-use graphs statically.

Instrumented code (by inserting logging function calls)
to collect dynamic data, such as common control flow
paths in dynamic (e.g., how frequently particular branch
is taken) and tracked dynamically allocated objects by
instrumenting memory allocation and usage. The
collected data was mapped into graph representation.

NOV 2024 – DEC 2024

Tools: C++, SFML

Graphics editor

GH link: <https://github.com/d3clane/MyPhotoshop>

Developed my own graphics editor as part of a student
group project, during which we discussed and
implemented standard API and developed various
plugins for it. Implemented plugins based on the
standard API, including a brush tool, eraser, Gaussian
blur filter, and brightness filter, among others.

Skills

- C
- C++
- Assembly x86-64
- Concurrency
- Graphviz
- Algorithms and data structures
- EDB
- Python
- SFML

Achievements

Russian National Physics Olympiad

Twice final stage silver medal
2022, 2023

Candidate to the national physics team for the international Olympiad

2022

International Experimental Physics Olympiad (IEPHO)

Bronze medal
2020

Russian National Informatics Olympiad

Twice region stage prize winner
2021, 2023

Languages

- Russian (Native)
- English (B2)

FEB 2025 - MAY 2025

Tools: C/C++

Concurrency course

1. Implemented synchronization primitives (mutex, condvar, spinlock, semaphore, waitgroup, etc.) using atomic operations only. Designed FIFO spinlock that avoids false sharing and the thundering herd problem.
2. Built a thread pool and coroutines. Combined both to create lightweight fibers.
3. Developed synchronization primitives for fibers (mutex, event, waitgroup, channel).
4. Implemented futures — a functional approach to concurrency modeled as graphs of transformations of immutable values.

As part of the course, work-stealing thread pool, lock-free data structures and other concurrency objects will be implemented in the future.

APR 2024

Tools: C/C++, ASM X86-64, kCachegrind, Python, Graphviz

Optimizing hash table using asm

GH link: <https://github.com/d3clane/HashTable>

Optimizing hash table requests time performance using asm. Three types of optimizations are used:

1. SIMD instructions
2. Inline asm
3. Linking with another file with asm code

FEB 2024

Tools: C/C++, Disasm, SDL

Hacking my friend's program (crackme)

GH link: <https://github.com/d3clane/DosHack>

Finding vulnerabilities in my friend's program and exploiting them.

Soft Skills

- Quick learning
- Organized
- Hard-working
- Sociable and collaborative
- Responsible